XRPL Validator — Daily Engineering Report · 2026-07-17

Period: 2026-07-17 00:00 to 24:00 US Eastern (2026-07-17 04:00 to 2026-07-18 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04); data.xrpl.org agreement scores. Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data.

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

External agreement, data.xrpl.org (UTC day 2026-07-17): 0.99995 (1 missed of 21,616). *data.xrpl.org scores UTC days from the validations its crawlers received.*

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **feat(node): parity_probe bin — runtime fixture replay for the scout loop** (`52e9df5` , xrpl-validator) Loads blobs+expected fixture files by path (no rebuild per ledger) and reports the standard parity block. Exit 0 clean / 1 divergent / 2 fixture error.
- **feat(node): network-parameterized replay — NetParams for testnet/devnet parity probes** (`8ed5545` , xrpl-validator) apply_ledger_in_order_with_net carries network_id/base_fee/reserves (mainnet wrapper keeps every existing caller unchanged); fixtures capture the params from server_state; parity_probe replays under the fixture's network rules.
- **feat(node): differential_probe — measure native engine vs mainnet parity** (`c862c37` , xrpl-validator) Replays each fixture tx through the native xrpl_ledger transactors and compares (TER, mutation-key-set) to mainnet — the instrument for the fully-Rust roadmap. Fixture generator now emits tx_json + tx_order for the native leg.
- **fix(node): differential_probe --json emits only JSON (SUMMARY to stderr); re-mint sample fixtures with tx_json** (`9cfb9af` , xrpl-validator)
- **fix(ledger): TrustSet decodes base58 issuer (was hex-only → every real tx temMALFORMED)** (`fdca115` , xrpl-validator) extract_limit_amount decoded the LimitAmount issuer with hex::decode, but mainnet JSON carries it as a base58 classic address — so every real TrustSet failed to Malformed before applying. Add decode_issuer (hex OR base58 via the canonical xrpl_core AccountId::from_address). Differential: TrustSet DIVERGE-TER 16->0 (all now tesSUCCESS matching mainnet, 1 full MATCH; the remaining 15 are DIVERGE-MUT — owner-directory …

- **feat(node): differential_probe loads pre-state objects (not just AccountRoots)** (`330ed4d`, xrpl-validator) Native was starved of pre-state — only AccountRoots were loaded, so it phantom-created every RippleState/Offer/Check/AMM object it referenced, inflating the whole conformance map. Now loads each tx's Modified/Deleted affected-nodes at seq-1 (transcoded to native's account-hex convention), so native SEES the objects it operates on. TrustSet lines now modify instead of phantom-create; the residual misses are genuine …
- **fix(ledger): NFTokenAcceptOffer reads offer ref as hex ledger-index, not {Account,Sequence}** (`31735dc`, xrpl-validator) NFTokenSellOffer/BuyOffer is the offer object's 32-byte hex ledger index; the native code treated it as an {Account,Sequence} pair and failed to Malformed before ever reading the offer. Now parses the hex index and takes the owner from the loaded offer object. Genuine bug fix; the transactor still has further naive parses (NFTokenID-as-object) + needs NFT-page transfer logic to green — noted, not chased here. …
- **feat(ledger): owner-directory bookkeeping (single root page) + wire into TrustSet** (`274d420`, xrpl-validator) New crate::ledger::directory: owner_dir_insert / owner_dir_remove maintain an account's owner directory (the DirectoryNode listing objects it owns). Shared infrastructure for every object-creating transactor. TrustSet now inserts a new RippleState into BOTH parties' directories (owner_dir_insert), so native emits the DirectoryNode mutations mainnet does — at the right key + kind (the gate compares key/kind sets, so …
- **feat: multi-page owner directory + load native-read keys — TrustSet 1->10 MATCH** (`a22af92`, xrpl-validator) Two increments that together get trust-line creates to byte-for-key parity: - directory.rs: owner_dir_insert now walks the page chain (root.IndexPrevious -> last page), appending to the last page or splitting to a new dir_page_key page on overflow — so native touches the SAME directory page mainnet does (large gateway issuers keep trust lines on non-root pages). - differential_probe: native_read_keys loads the …
- **feat(xrpl-ledger): trust-line deletion + oracle state threading — TrustSet 14/14 MATCH** (`87d6ff7`, xrpl-validator)
- trust_set: rippled-faithful default-line deletion (per-side interest test: limit/balance/quality/freeze/auth/NoRipple-vs-DefaultRipple), tf-flag application (tfSet/ClearNoRipple, tfSet/ClearFreeze) before the test, creation flags (creator reserve bit + auto-NoRipple), both-roots OwnerCount touch at delete mirroring create - directory: hint-based owner_dir_remove (rippled's no-walk LowNode/HighNode design) with …
- **feat(xrpl-ledger): OfferCancel directory unlinking — 6/6 MATCH across two ledgers** (`270345a`, xrpl-validator)
- directory: generalize dir_remove(root_key, ...) so order-book quality dirs share the owner-dir page mechanics; owner_dir_remove is now a wrapper. Hint path no longer requires the root page loaded (book hints often ARE the root). Empty-root cascade: deleting the only non-root page kills the whole directory when the root holds no entries (rippled behavior). - directory: hex-case normalization — mainnet-loaded …

Tests and campaigns

- **test(node): arm scout find #105663160 — NFTokenAcceptOffer silent-TER parity gate** (`3bac75a`, xrpl-validator)
- **test(node): 5 coverage fixtures (AMM/Check/NFToken-accept) for engine-wide conformance map** (`f70baf9`, xrpl-validator)

Docs and operations

- **revert NFToken offer-ref parse (needs focused pass); keep AMM XRP+base58 encoding fix** (`51c9665`, xrpl-validator)

Independence program

No independence record for UTC day 2026-07-17; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Validator Bonding With Slashing-Backed Finality** (consensus) Add a new owner-reserved SLE `ltVALIDATOR_BOND` (keylet::validatorBond(nodePublicKey)) holding sfAccount, sfValidatorPublicKey, sfBondAmount (escrowed
- **Concentrated Liquidity Ranges for AMM Pools** (amm-defi) Add a new owner-reserved SLE `ltAMM_POSITION` (keylet::ammPosition(ammID, account, sequence)) carrying sfTickLower, sfTickUpper (fixed-point ticks on
- **Domain-Gated Eligibility for MPT Transfers** (rwa) Add sfDomainID (the existing PermissionedDomain ID type already used by PermissionedDEX) to `MPTokenIssuance`, settable via a new `tfRequireDomain` fl

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
07-17 15:07	—	xrpl-validator	`52e9df5`	feat(node): parity_probe bin — runtime fixture replay for the scout loop
07-17 15:57	—	xrpl-validator	`3bac75a`	test(node): arm scout find #105663160 — NFTokenAcceptOffer silent-TER parity gate
07-17 17:04	—	xrpl-validator	`8ed5545`	feat(node): network-parameterized replay — NetParams for testnet/devnet parity probes
07-17 19:34	—	xrpl-validator	`c862c37`	feat(node): differential_probe — measure native engine vs mainnet parity
07-17 19:40	—	xrpl-validator	`9cfb9af`	fix(node): differential_probe --json emits only JSON (SUMMARY to stderr); re-mint sample fixtures with tx_json
07-17 20:16	—	xrpl-validator	`f70baf9`	test(node): 5 coverage fixtures (AMM/Check/NFToken-accept) for engine-wide conformance map
07-17 20:27	—	xrpl-validator	`fdca115`	fix(ledger): TrustSet decodes base58 issuer (was hex-only → every real tx temMALFORMED)
07-17 20:43	—	xrpl-validator	`330ed4d`	feat(node): differential_probe loads pre-state objects (not just AccountRoots)
07-17 20:47	—	xrpl-validator	`31735dc`	fix(ledger): NFTokenAcceptOffer reads offer ref as hex ledger-index, not {Account,Sequence}
07-17 21:11	—	xrpl-validator	`51c9665`	revert NFToken offer-ref parse (needs focused pass); keep AMM XRP+base58 encoding fix
07-17 21:29	—	xrpl-validator	`274d420`	feat(ledger): owner-directory bookkeeping (single root page) + wire into TrustSet
07-17 21:43	—	xrpl-validator	`a22af92`	feat: multi-page owner directory + load native-read keys — TrustSet 1->10 MATCH
07-17 23:20	—	xrpl-validator	`87d6ff7`	feat(xrpl-ledger): trust-line deletion + oracle state threading — TrustSet 14/14 MATCH
07-17 23:50	—	xrpl-validator	`270345a`	feat(xrpl-ledger): OfferCancel directory unlinking — 6/6 MATCH across two ledgers

Times in UTC. A commit appears on the day it was committed to main or merged into it.